

Patrick R. Kothbauer

Artificial Intelligence Engineer

I'm a native swede, enjoying life in beautiful Sweden

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Professional Summary

I'm a trained Computational Linguist, specializing in Large Language Models (LLM) and Agentic Artificial Intelligence. I'm passionate about building systems that are simple, have few parts, are easy to maintain, and very fast. I'm looking for my first professional developer job. I have a couple of years of experience in developing Open Source Software.

Education

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| Paused - | Chalmers and the University of Gothenburg, Sweden
<i>Currently enrolled in the Bachelor of Science programme in Mathematics.</i> |
| 2004 - 2007 | University of Gothenburg, Sweden
<i>B.Sc. Computational Linguistics. Coursework included Logic, Prolog, Formal Syntax, Formal Semantics, Phonetics, Dialogue Systems, Programming for Corpus Linguistics, Artificial Neural Networks, Research Methodology, Automatic Analysis of Speech and Speech Synthesis, and Parsing Algorithms</i> |
| 1996 - 1997 | Prosolvia Interactive, Sweden
<i>Virtual Reality Trainee. Coursework included Programming in Java, 3D modelling with Pro/Engineer, Adobe Photoshop, Illustrator, Premiere, Macromedia Director, Sound Editing, and VR simulation using EON Reality.</i> |

Experience

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| 2018 - 2025 | Financial Markets and Blockchain Developer
<i>Freelance Developer</i>
For close to 7 years I have been programming open source software for blockchain and financial markets. Side projects that I have worked on include: HoneyLedger - a blockchain that can be used to track honey from producer to consumer shelf, BikeChain - a proof of concept where bike owners can register their bike on a blockchain to prove ownership, MiniFIX - a Financial Exchange Information Rust crate, PineCoder - developed hundreds of financial indicators and trading strategies in the PineScript language, that can be used on the TradingView platform. I'm currently exploring Vibe-Coding as a method for rapid prototyping of software. |
| 2015 - 2018 | Swedish Chamber of Digital Currency Commerce
<i>Founder</i>
Founded the non-profit association SCDCC (Swedish Chamber of Digital Currency Commerce). The purpose of the association was to increase knowledge about Bitcoin, digital currencies, and blockchain technology among companies, consumers, and government authorities. Assisted with marketing work for the founders of the cryptocurrency Ethereum prior to its launch. Met with Swedish minister of digitalisation, and organised the Blockchain for Banks event. Organized multiple Blockchain events under the BitCoast name. |
| 2009 - 2014 | Bäck-n-Bauer HB and Lucky Punk AB
<i>Co-Founder (BnB), Producer, and Director</i>
Worked with corporate clients, such as *Bic, Volvo AB, Volvo Construction Equipment, Telia, and Estrella,* in producing and directing marketing commercials. Gained solid experience in planning and executing a complete ad commercial production, leading a team, managing multiple projects, editing video, recording audio, and working with special effects teams. Gained broad experience in producing audiovisual content, from initial client pitch to final delivery. |

2007 - 2009	Semcon Human-Machine Interface Division <i>Design Engineer & Project Co-coordinator</i> Developed proof-of-concepts in Human-Machine Interaction and ergonomics. Was assigned as responsible for language technology area at Semcon HMI. Gave lectures in the fields of Artificial Intelligence and Dialogue Systems. Programmed multimedia marketing in Adobe Flash for Semcon AB. Co-coordinator (together with Patrik Palo, head of HMI-development at Volvo Car Corporation) for Pre-EFESOS, an 8 month research project on next-generation in-vehicle Human-Machine Interaction .The project resulted in approved funding of over 80 million SEK from VINNOVA.Participated in the development of research proposals. Coordinated budgeting work among participants from Chalmers, Volvo Car Corporation, Lund University, HiQ, and the Viktoria Institute.
1998 - 2004	Prosolvia Research & Technology AB and E3 Communication <i>Web Developer and Graphic Designer</i> I was responsible for developing the company's website, primarily using HTML for all design and layout since CSS1 was not yet fully supported by major browsers such as Netscape and Internet Explorer. Under the guidance of the Marketing Manager, I also created marketing materials for both web and print, mainly utilizing QuarkXpress and Adobe Pagemaker. Additionally, I assisted engineers with a range of graphic production tasks.

Open Source Software Development and Portfolio Projects

1. 100DaysOfVibe: For a period of 100 days, I'm vibe-coding a project every day.
Project Website: <https://100-days-of-vibe.vercel.app/>
2. Rust crates: I develop and maintain the following Rust open source libraries.
Crates Link: <https://crates.io/users/cryptopatrck>
3. VesselWise: LLM-chat interface over shipping IoT-data.
GitHub Link: <https://www.github.com/cryptopatrck/vesselwise>
4. BallerMentality: Chat with basketball 2D game characters (Agents), built with LangGraph.
GitHub Link: <https://www.github.com/cryptopatrck/ballermentality>

Social Media Links

1. Twitter: <https://www.x.com/cryptopatrck>
2. LinkedIn: <https://www.linkedin.com/in/cryptopatrck>
3. Instagram: <https://www.instagram.com/cryptopatrckk>

Community and Communication

1. Co-organizer: of the Unofficial Libra Blockchain Hackathon 2019, where programmers around the world competed to develop applications using Facebook Libra's cryptocurrency, Libra. The prize was 1 Bitcoin (today worth over 1,000,000 SEK). Link: <https://medium.com/@hacklibra/hacklibra-winners-8da1a9124628>
2. Keynote Speaker: at SEB HQ (Stockholm) for the Capital Market Experts annual meeting, 2015. Gave a one-hour presentation on Bitcoin. The crowd roared with laughter when I claimed that the price of a single bitcoin would one day reach well above 10,000 dollars. Link: <https://www.nea.nu/webinar-bitcoin-vad-ar-det-och-hur-fungerar-det/>
3. Project Leader: Assisted Charlie Lee, the creator of the Litecoin cryptocurrency during a trademark infringement case. The dispute was resolved by the EU Patent Office in Spain. Link: <https://x.com/LTCFoundation/status/710425904602861568>

Writings

1. Agentic Micro Economy (2025) Coordination of work between AI Agents by spinning up an Auction to bid on Requests-for-Work contracts. The contract winner buys Prompt-run from LLM MCP Server. Payment between User, Agent, and Server is settled on blockchain.
2. Reinforcement Learning Combat Strategies for Weak Agents (2025) Using RL to discover mathematically proven edge in 1v1 or n-v1 combat situations.
3. All good things... (2025). A graph-algorithm that attempts to ballance the effect of 'Winner takes it all' patterns that often emerge in online expert oriented services with long tails.
4. Block Time Lock (2024). Technical post of a cryptographic protocol for sending digital messages into the approximate future with probabilistic security guarantees provided by the Bitcoin blockchain.
5. Kothbauer, P. (2008). Thesis: Ontologies in Issue-based Dialogue Systems. Thesis work for B.Sc. in Computational Linguistics - awaiting publication.

Skillset

1. AI: LLM development from scratch, OpenAI API, MCP, LangChain, LangGraph, RAG, Qdrant Vector DB, Vercel AI SDK, LM Studio, evaluation and fine-tuning.
2. Programming: Rust, Python, TypeScript, C, Scala, Prolog, PyTorch, JAX, MLX, Wasm.
3. Development: Linux, PowerShell, Git, Docker, Kubernetes, TerraForm, GCP, Figma, React, Tailwind, FastAPI, SQLAlchemy, Axum, Postgres, Redis, OpenAPI, Docusaurus.

Fun Facts

1. In 2016, I helped start the political party Demokraterna together with Gothenburg politician Martin Wannholt. I was part of the original steering group that met at the City Hall in Gothenburg, Sweden. One achievement I am particularly proud of, is that I arranged for the party to move from its original web address, www.demokraternaigoteborg.se to www.demokraterna.se. The later domain was owned by a politician in the Stockholm area. I was able to contact him via the domain Internet Service Provider and successfully negotiate the purchase of <https://demokraterna.se/>. I think that showed initiative from my side.
2. With a group of friends, I organized Klubb Locus (a deep house club night at the venue Pustervik in Gothenburg) every month for 20 years in a row. This experience taught me a lot about organization events, communication, consistent branding and marketing.